

Name: \_\_\_\_\_

Date: \_\_\_\_\_

# PERCENTAGES OF AMOUNTS, (PERCENTAGE INCREASE AND DECREASE)

**LO: I can solve problems involving percentage increase and decrease of amounts.**

## Increasing and decreasing

To **increase** by a percentage: find the percentage of the amount and **add** it to the original. To **decrease**: find the percentage and **subtract**.

**Quick method:** Use a single calculation. Increase by 20% multiply by 1.2. Decrease by 20% multiply by 0.8.

## Quick examples

|   |                              |                        |
|---|------------------------------|------------------------|
| ● | £50 increased by 10% = £55   | add 10% of 50          |
| ● | £80 decreased by 25% = £60   | subtract 25% of 80     |
| ● | £120 increased by 50% = £180 | add half               |
| ● | £50 increased by 10% = £60   | 10% of 50 is 5, not 10 |
| ● | £80 decreased by 25% = £55   | 25% of 80 is 20        |

## Key idea

Find the percentage, then add (increase) or subtract (decrease) from the original.

**Increase:** **Original + %** | **Decrease:**

**Original - %**

Or use the single multiplier method

## MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

### Example 1 - increase (add)

Increase £60 by 15%.

$$\begin{aligned} 15\% \text{ of } 60: 10\% &= 6, 5\% = 3 \\ 15\% &= 6 + 3 = 9 \\ 60 + 9 &= £69 \end{aligned}$$

**£69**

Find 15% then add it to the original.

### Example 2 - decrease (subtract)

Decrease £240 by 35%.

$$\begin{aligned} 10\% \text{ of } 240 &= 24 \\ 30\% &= 72; 5\% = 12; 35\% = 84 \\ 240 - 84 &= £156 \end{aligned}$$

**£156**

Find 35% then subtract from the original.

### Example 3 - single multiplier increase

A £450 TV increases by 8%. Find the new price.

$$\begin{aligned} \text{Multiplier} &= 1.08 \\ 450 \times 1.08 \\ &= £486 \end{aligned}$$

**£486**

Multiply by 1.08 in one step for speed.

### Example 4 - single multiplier decrease

A £320 bike is reduced by 12.5% in a sale.

$$\begin{aligned} \text{Multiplier} &= 1 - 0.125 = 0.875 \\ 320 \times 0.875 \\ &= £280 \end{aligned}$$

**£280**

Multiplier = 1 - 0.125 = 0.875.

## YOUR TURN – PRACTICE (deliberate practice)

### 1. Percentage increase (add method)

Increase each amount by the given percentage.

- a) £40 by 10%
- b) £70 by 20%
- c) £150 by 5%
- d) £300 by 30%

### 2. Percentage decrease (subtract method)

Decrease each amount by the given percentage.

- a) £80 by 25%
- b) £200 by 15%
- c) £360 by 10%
- d) £50 by 40%

### 3. Multiplier method

Use a single multiplier to calculate.

- a) £250 increased by 12%
- b) £480 decreased by 15%
- c) £90 increased by 45%
- d) £600 decreased by 22%

### 4. Word problems

Solve each problem.

- a) A £25 pizza deal increases by 8%. New price?
- b) A 500 ml bottle is 20% bigger. New volume?
- c) A £180 coat has 35% off. Sale price?
- d) Bus fare of £2.40 rises 5%. New fare?

### 5. Mixed increase and decrease

Calculate the final amount.

- a) £95 increased by 20%      b) £450 decreased by 18%      c) A town of 8000 grows by 6%      d) £75 reduced by 12%      e) A £1200 rent increases by 2.5%

## COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) Increasing £60 by 10% gives £66 ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

b) Decreasing £100 by 25% gives £125 ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

c) The multiplier for a 12% increase is 1.12 ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

d) The multiplier for a 30% decrease is 1.30 ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

e) 50% decrease means halving the amount ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

## 6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A shop increases prices by **10%** in January, then decreases them by **10%** in the sale. A customer says the price is back to normal. Show with a **£200** item why this is incorrect.

Expression: \_\_\_\_\_

Simplified: \_\_\_\_\_

b) A football stadium holds **28,000** people. An extension increases capacity by **15%**. Safety rules then reduce the new capacity by **5%**. What is the final capacity?

Expression: \_\_\_\_\_

Simplified: \_\_\_\_\_

### TOP TIPS

Read the question carefully. Show all your working step by step.  
Check your answer makes sense.

### CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

